Grow-Light Pulley System
Stress Analysis

PHS Sustainability Capstone

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ME486C

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Steve Sanghi College of Engineering

I. Introduction

The goal of this analysis is to provide a full design and stress analysis for the grow-light pulley system at the Ponderosa High School (PHS) greenhouse. The grow-light installation will improve plant growth for PHS staff and students. Additionally, learning how to garden, maintain a healthy environment for plants, and working on a team for plant growth teaches PHS students' valuable soft skills for post-graduation. This document will showcase the proposed design for the grow-light pulley system with possible variations that the clients, John Taylor and Les Haurer, may prefer. A stress analysis will be conducted on the system with the max loads the system will encounter. Through this analysis, the team will have a conclusive, validated design for installation at the PHS greenhouse.

II. Assumptions and Justifications

Provided below are the assumptions and justifications for the grow-light pulley system. Each assumption is given sufficient justification to describe why the assumption is permitted. These assumptions will be used throughout the stress analysis.

- ***Assumption 1:*** The weight of the system (grow-light casing and grow-lights) is estimated to be 12.6 kg (50.4 kg for all grow-light casings and lights)
  Justification: Estimated grow-light casings and lights are found from manufacturer specifications in the Appendix. Although the grow light casings and lights are not the exact model, they are in similar size and weight to the manufacturer specifications. Any differences in weights are small enough to be negligible.

- ***Assumption 2:*** The rope and pulley system have negligible friction
  Justification: There will be a small amount of friction due to real world applications but to simplify calculations, it is assumed that the contact is frictionless.

- ***Assumption 3:*** The rope and chain attached to the grow-lights have a negligible weight
  Justification: In comparison to the weight of the grow-light casings and lights, the rope and chain are adding very little weight and therefore is negligible.

- ***Assumption 4:*** All calculations can be performed at STP
  Justification: The grow-light pulley system will be installed in Flagstaff, AZ which is 7000ft above sea level. This results in different temperature and pressure values but overall is a small difference and therefore is negligible.

- ***Assumption 5:*** The chain makes an equilateral triangle with the grow-light casing
  Justification: As seen in the physical modeling section, there is a triangle made of chain with the grow-light casing. The eye screws in the grow lights are presumed to be equidistant to the center with minor differences in length. This means that the triangle will be almost exactly an equilateral triangle.

- ***Assumption 6:*** The PHS greenhouse ceiling is made from standard sheet metal with wooden support beams
  Justification: The team does not have information on the specific sheet metal or wooden support beams used for the ceiling of the PHS greenhouse. So, it is assumed that standard sheet metal and 2x10" [1] wooden support beams used for manufacture considering that the greenhouse was built on a tight budget.

- ***Assumption 7:*** The allowable bending stress of the support beams is 10.342 MPa [2]
  Justification: Validated with outside source for builders and homeowners.

III. Details of Physical Modeling

In this section, the physical modeling used to develop equilibrium equations, material selection, stress calculations, and SolidWorks modeling are shown below. Variations of how the grow-light pulley system is set up so the team can adjust according to the client's wants. The main adjustment between designs is whether the grow-lights move in sets of one, two, or four. Being able to adjust each grow-light system gives the clients more flexibility over which plants can be grown, when they can start growing, and how many species of plants, with different light needs, can grow at a time. But it is important to consider that having individual movement for each grow-light system will result in more time and labor when the system needs to be moved.

After confirming that the system will appropriately hold the grow light pulley system with the factor of safety accounted for, the team may need to do further analysis for the chosen design by the clients.

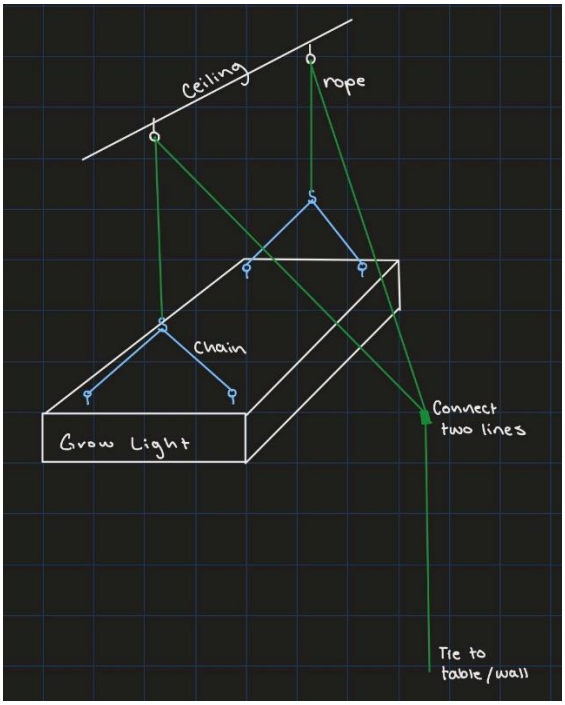


Fig 1. Individual grow-light pulley system

The figure above shows an individual grow-light system. The grow light casing would be attached to the ceiling by using chain that runs through 4 eye screws on the casing. The chain is then connected to rope that runs through eye screws on the ceiling. The two ropes are connected, and an anchor place will be attached to the table or wall to keep the grow-lights at the preferred height.

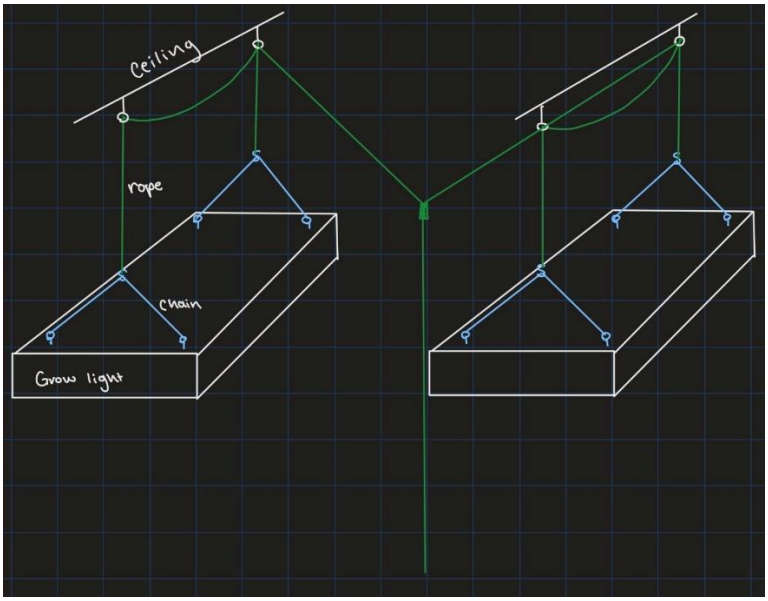


Fig 2. Set of 2 grow-light pulley systems

Figure 2 shows a set of 2 grow-light systems. This variation would allow for the grow-light pulley systems to move simultaneously with each other.

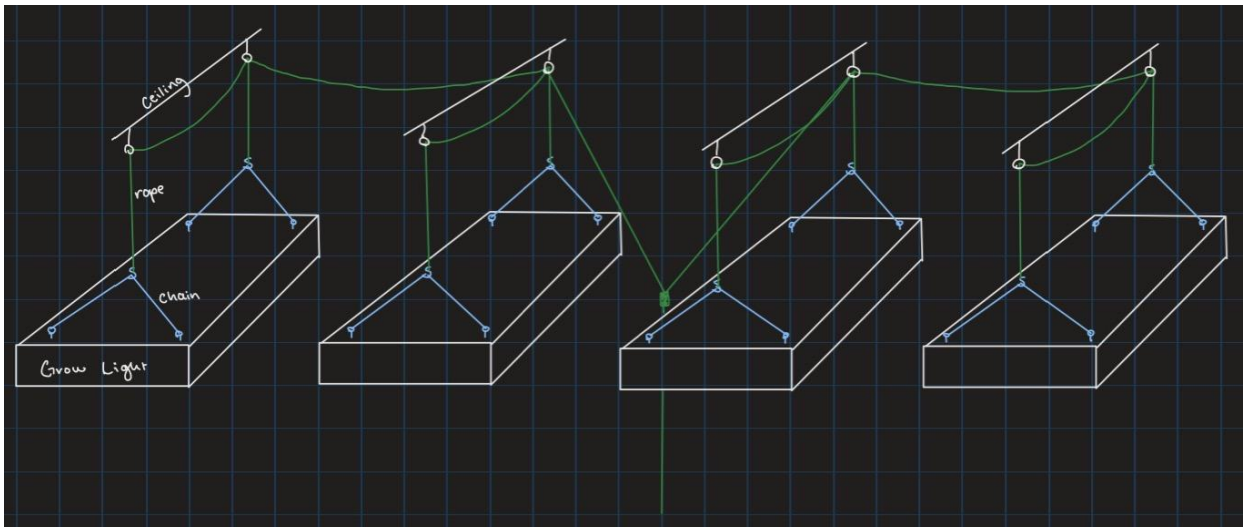


Fig 3. Set of 4 grow-light pulley systems

Figure 3 shows a set of 4 grow-light systems. This variation allows for all four grow-light pulley systems to move simultaneously with each other.

Note about physical modeling: Future modeling and FEA analysis will be conducted when the updated SolidWorks student licenses are granted to the team.

IV. Equations

In this section, a force and stress analysis is conducted to confirm that the grow-light pulley system will be structurally sound when installed into the PHS greenhouse. In table I and II below, all symbols, equations, and units are identified for reference throughout the force and stress analysis.

Table I

Symbols used for Force and Stress Analysis

Symbol	***Description***
F_y	Forces in the 'y' direction
F_x	Forces in the 'x' direction
M_o	Moment around 'o' point
M_{max}	Maximum moment on a beam
σ	Bending stress
σ'	Allowable bending stress
c	Centroid of a beam
I	Inertia
FS	Factor of safety

Table II

Equations/units used for Force and Stress Analysis

Equations [3]	***Units***
$+\uparrow \Sigma F_y = 0$	N
$+\rightarrow \Sigma F_x = 0$	N
$+\circlearrowleft M_o = 0$	N*m
$\sigma = \frac{M_{max} * c}{I}$	Pa
$FS = \frac{\sigma'}{\sigma}$	-

To begin the stress analysis on the grow-light system, a force analysis was generated to understand what would be on the wooden support beam. Below are the free body diagrams for one of the rope/chain holders.

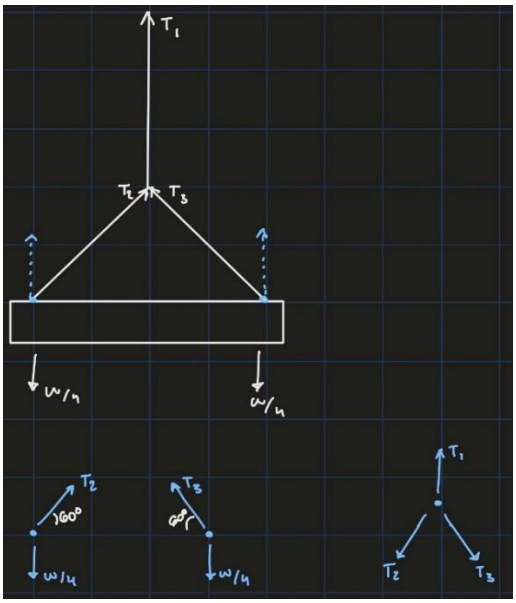


Fig 4. Forces from grow-light system

Using the equilibrium equations for the 'y' and 'x' direction, the force of T_{2y} and T_{3y} is **30.90N**. This means the overall tension being placed on the support beam (T_1) is **61.80N**. These forces can now be translated to loads on the wooden support beam in a stress analysis.

Below is a diagram of the loads from one grow-light pulley system. There are two point loads for each system since the grow-light system is connected to the ceiling with two ropes and eye screws.

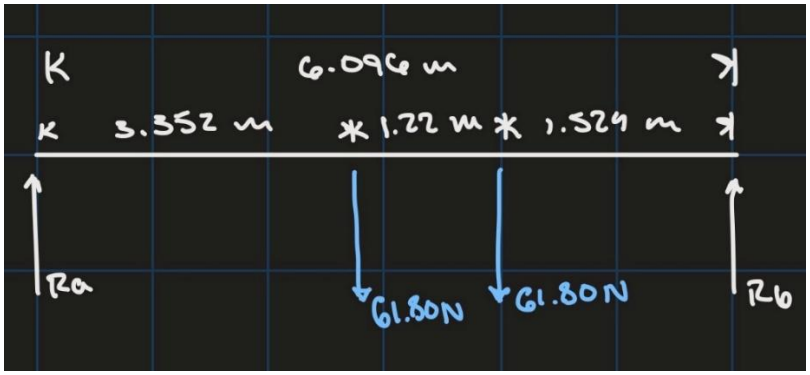


Fig 5. Load diagram for grow-light pulley system

Using this diagram and equilibrium equations for force in the 'y' direction and moment in the 'z' direction, the reaction forces for R_a and R_b are **43.28N** and **80.32N**, respectively. This information allows for a shear and momentum graph to be generated.

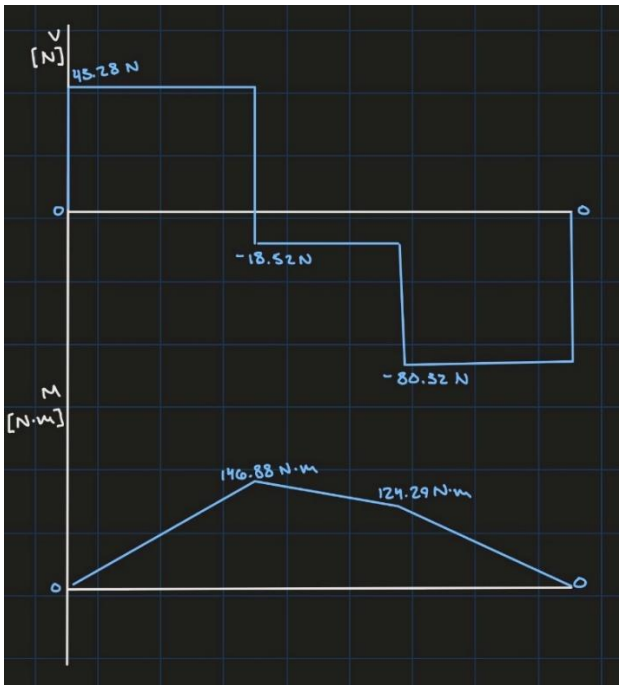


Fig 6. Shear and moment graph for grow-light pulley system

These graphs show that the maximum moment on the beam will be **146.88 N*m**. This is where the bending stress calculation will be found because this is the worst-case scenario for the support beam.

V. Results

From the force and stress analysis, the total bending stress on the wooden support beam for the grow-light pulley system is **0.6436 MPa**. The bending stress is found with the assumption that the support beam's cross section is 2x10". According to [2], the recommended allowable bending stress for ceiling support beams is **10.342 MP**a (Assumption 7). This means that installing the grow-light pulley system will be structurally safe with the factor of safety being **16**. Using this information, the team has validated the grow-light pulley system is safe to install for clients. Additional calculations/analysis could be from FEA analysis with SolidWorks modeling and further analysis on material selection for rope and chain in the grow-light system.

VI. Conclusion

This document contains the components for the stress analysis on the grow-light pulley system. All assumption and justifications for the assumptions are defined to start the stress analysis. Design variations are shown for client preference and to add possible adjustments for the team during installation. A force analysis is conducted to find the main tension forces that will be applied onto the wooden support beam. Then a stress analysis finalizes the bending stress and with validation from online sources, confirms the grow-light pulley system for installation at the PHS greenhouse. With this analysis, the team can move forward with manufacturing.

References

[1] Home Depot, Types of lumber, https://www.homedepot.com/c/ab/types-of-lumber/9ba683603be9fa5395fab90567851db.

[2] “Structural Wood Beams Guide for Builders and Homeowners,” *Sierralogandtimber.com*, 2026. Available: https://sierralogandtimber.com/structural-wood-beams/.

[3] B.C. Hibbeler, *Mechanics of Materials*, 10th ed. Hoboken, NJ, USA: Pearson, 2015

Appendix

Product Information Bulletin

GRO-LUX® Aquarium and Aquarium Wide Spectrum

Fluorescent Lamps for indoor gardening



- **Fluorescent lamps for indoor gardening**

  Provide essential energy for
  - Germination, propagation and growth
  - Photosynthesis
  - Chlorophyll synthesis
  - Enhancing vegetation
- **Choice of Lamps**
  - GRO-LUX Aquarium
    for plants requiring indirect sunlight
  - GRO-LUX Aquarium Wide Spectrum
    for plants requiring full sunlight
  - Combination of GRO-LUX Aquarium
    and GRO-LUX Aquarium Wide Spectrum
    were appropriate

SYLVANIA's GRO-LUX Aquarium and GRO-LUX Aquarium Wide Spectrum fluorescent lamps provide the essential energy for use in indoor gardening applications. Light provides the energy that plants need to produce food and other substances required for growth and flowering. These lamps provide the types of light that are most beneficial for these essential processes.

GRO-LUX Aquarium lamps provide the right balance of red and blue wavelengths for low-energy plants such as house plants. High-energy plants such as vegetables seem to require more energy in the red portion of the spectrum. The GRO-LUX Aquarium Wide Spectrum lamps or a 50/50 mix of the two types of GRO-LUX lamps may produce the best results for the high-energy plants.

In addition to providing essential energy for growth and maintenance, GRO-LUX lamps bring out the vibrant natural colors of the foliage, flowers and vegetables enhancing their appearance.

Product Availability

Wattage	Ordering Description
15	F15T8/GRO/AQ/RP
15	F15T12/GRO/AQ/WS/RP
20	F20T12/GRO/AQ/RP
20	F20T12/GRO/AQ/WS/RP
30	F30T8/GRO/AQ/RP
40	F40T12/GRO/AQ/RP
40	F40T12/GRO/AQ/WS/RP

Application Information

Applications

For additional information, please call 1-800-LIGHTBULB or visit www.sylvania.com to obtain a copy of the Technical Information Bulletin "Light and Plants".

Fixtures

Contact your local fixture agent for available fixtures.

Ballast Information

Contact your OSRAM SYLVANIA representative for a list of compatible ballasts for GRO-LUX lamps.

Application Notes

1. The F15T8, F20T12 & F30T8 lamps are preheat lamps and operate on the same ballasts and require the same starters as other preheat lamps of the same wattages. The lamps may also be operated on the appropriate trigger start ballasts.
2. The F40T12 lamps are rapid start lamps and operate on the same ballasts as other F40 rapid start lamps.
3. Low-energy plants require about 15 fluorescent lamp watts per square foot about 12-15 inches above the plants.
4. High-energy plants require at least 20 fluorescent lamp watts per square foot about 12-15 inches above the plants.
5. Germinating seeds or cuttings being rooted requires about 10 lamp watts per square foot with the lamps about 6-8 inches above the plants.
6. For photosynthesis, if using incandescent lamps, you would need 2½ to 3 times the fluorescent wattage to provide PAR values equal to GRO-LUX Aquarium or GRO-LUX Aquarium Wide Spectrum lamps.
7. For equal light, if using incandescent lamps, you would need 1½ to 2 times the fluorescent wattage to provide light equal to the GRO-LUX Aquarium lamps and 2 to 2½ times the fluorescent wattage to provide light equal to the GRO-LUX Aquarium Wide Spectrum lamps.

SYLVANIA

FL039

Fig A1. Grow-light specifications sheet 1

Sample Specification
Lamp(s) shall be (GRO-LUX/Aquarium or GRO-LUX/ Aquarium/Wide Spectrum) lamp(s). Lamps shall provide the red and blue wavelengths of energy needed for photosynthesis, and chlorophyll synthesis in plants and in general, provide the light needed for plants to flourish and look their best.

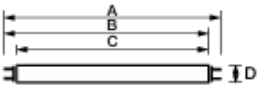
Ordering and Specification Information

Item Number	Ordering Abbreviation	Watts	Bulb	Base	Length	Avg. Rated Life (hrs.)	Initial Lumens	CCT	CRI
21657	F15T8/GRO/AQ/RP	15	T8	Medium Bi-Pin	18"	7500	325	N/A	N/A
22029	F20T12/GRO/AQ/RP	20	T12	Medium Bi-Pin	24"	9000	480	N/A	N/A
23160	F30T8/GRO/AQ/RP	30	T8	Medium Bi-Pin	36"	7500	800	N/A	N/A
24660	F40T12/GRO/AQ/RP	40	T12	Medium Bi-Pin	48"	20,000	1200	N/A	N/A
22013	F20T12/GRO/AQ/WS/RP	20	T12	Medium Bi-Pin	24"	9000	750	3400K	89
24671	F40T12/GRO/AQ/WS/RP	40	T12	Medium Bi-Pin	48"	20,000	1875	3400K	89

Ordering Guide

F	40	T12	/	GRO	/	AQ	/	WS	/	RP
Fluorescent	Wattage: 15, 20, 30 or 40 watts	Tubular glass 12/8" (1.5") diameter		GRO-LUX		Aquarium		Wide Spectrum		Retail Package

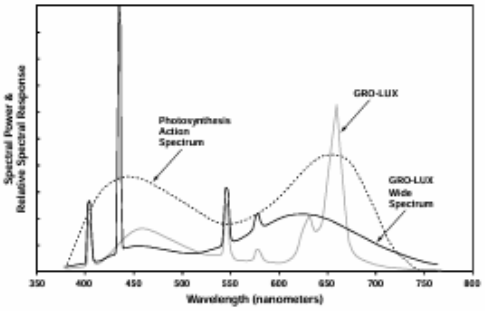
Dimensions



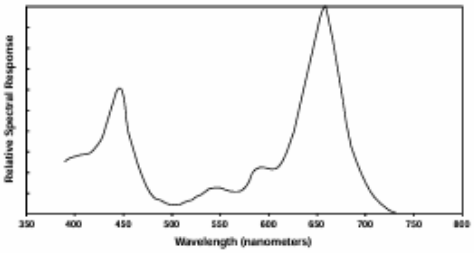
	(A) MOL (in)	(B)	(C)	(D) Diameter (in)
F15T8	17.91"	17.50"	17.22"	1.10"
F20T12	23.78"	23.50"	23.22"	1.59"
F30T8	35.78"	35.50"	35.22"	1.10"
F40T12	47.78"	47.50"	47.22"	1.59"

Technical Information

Photosynthesis and GRO-LUX Lamps



Action Spectrum for Chlorophyll Synthesis



OSRAM SYLVANIA
National Customer Support Center
18725 N. Union Street
Westfield, IN 46074

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Fax: 1-800-255-5043

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Fax: 1-800-562-4674

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Fax: 1-800-762-7192

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Fax: 1-800-762-7192

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Fig A2. Grow-light specifications sheet 2

Catalog # : _____ Project : _____ Type : _____

Prepared By : _____ Date : _____

Peak Series High Bay (PSHB)

Industrial



OVERVIEW	
Lumen Package (lm)	12,000 - 72,000
Wattage Range (W)	68 - 405
Efficacy Range (LPW)	177 - 189
Weight lbs (kg)	9.6 - 27.8 (4.3 - 12.6)
Control Options	OMSx, IMSBTxL, ALBC(S), ALSC(S)

QUICK LINKS

Ordering Guide | Performance | Photometrics | Dimensions

FEATURES & SPECIFICATIONS

Construction

- Rigid housing is die-formed steel.
- All metal parts are painted after fabrication following treatment with phosphate rust inhibitor. Finish coating of housing reflecting surfaces is with high-reflectance (minimum 92%) polyester powder.
- Standard finish is gloss white. Custom color available as special request.
- Quick wire access panel allows for easy power connections.
- Manufactured in the USA.

Optical System

- Available with clear or diffuse flat acrylic lens, clear or diffuse curved acrylic lens, or without lens for maximum light output.
- For protection in high impact environments, there is an optional clear or diffuse flat polycarbonate lens available.
- Optional narrow optic available for optimal aisle distributional
- All LED boards are rotatable in 30 degree increments from 0 (direct downlight) to 180 (complete uplight) for maximum versatility in the illuminated space (see product dimensions section for details).
- Available in 4000K, 5000K CCT. Contact factory for other CCT options.
- Minimum CRI of 80.

Electrical

- Standard Universal Voltage (120-277 VAC) Input 50/60Hz or optional High Voltage (347-480 VAC).
- L70 Calculated Life: 100,000+ hours.
- High Efficiency Driver: minimum 85%.
- 0-10V dimming (1% - 100%) standard.
- Optional 120V-480V external emergency battery pack is available to meet critical life safety lighting requirements. The 90-minute batteries provide constant power to the LED system, ensuring code compliance. A test switch/indicator button is installed on the housing for ease of maintenance.
- Emergency battery backup available up to 30W output (see EM lumen output table for details).
- Operating temperature range -40 to +60C (-40F to +141F) refer to ambient temperature table for specific units.
- Standard 6kV surge protection
  - 10kV surge protection optional.
  - All surge protection devices meet minimum Category C Low operation (per ANSI/IEEE C62.41.2).

Controls

- Controls Options Include:
  - Simple Sensor PIR end mount motion sensor available (OMSx).
  - Basic Bluetooth integral passive infrared sensor (IMSBTxL) with bluetooth app available.
  - Airlink Blue integral controller (ALBC) or integral controller with motion & photo sensor (ALBCSx).
  - LSI's AirLink Synapse wireless control system options reduce energy and maintenance costs. To order select ALSC or ALSCS (see ordering guide) as the controls option.

Installation

- Mounting options include: hook cable mount (MCH), toggle cable mount (TCH - see application note below), pendant mount (PMK) or surface mount (SMK). See accessory table for appropriate mounting hardware ordering information.
  - For EM applications, it is highly recommended to choose the TCH cable mount option to allow for side-to-side adjustment when countering the weight of the external battery (EM/EMHV).
- To maintain ambient rating, installation height can not be lower than 12'.

Warranty

- LSI luminaires carry a 5-year limited warranty. Refer to https://www.lsicorp.com/resources/terms-conditions-warranty/ for more information.

Listings

- Listed to UL 1598 and UL 8750.
- Meets Buy American Act requirements.
- Title 24 Compliant; see local ordinance for qualification information.
- Suitable for damp locations.
- IK10 rated with polycarbonate lenses per IEC 66262 mechanical impact code.
- DesignLights Consortium® (DLC) qualified product. Not all versions of this product may be DLC qualified. Please check the DLC Qualified Products List at www.designlights.org/QPL to confirm which versions are qualified.
- Specifications and dimensions subject to change without notice.

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Fig A3. Grow-light casing specifications sheet 1

Peak Series High Bay (PSHB) Industrial

Type : _____

Have questions? Call us at (800) 436-7800

ORDERING GUIDE

Back to Quick Links

TYPICAL ORDER EXAMPLE: PSHB 2 24L FAC D UNV 840 OMS2 EM SP1

Prefix	Length	Lumen Package	Lens/Distribution	Optical Presets	Voltage	Color Rendering/Temperature	Finish
PSHB - Peak Series High Bay	**2** - 2ft **4** - 4ft	**2 Foot** **12L** - 12,000 Lumens (2 row) **15L** - 15,000 Lumens (2 row) **18L** - 18,000 Lumens (2 row) **24L** - 24,000 Lumens (4 row) **30L** - 30,000 Lumens (4 row) **36L** - 36,000 Lumens (4 row) **4 Foot** **30L** - 30,000 Lumens (2 row) **36L** - 36,000 Lumens (2 row) **48L** - 48,000 Lumens[2] (4 row) **60L** - 60,000 Lumens[2] (4 row) **72L** - 72,000 Lumens[2] (4 row) Custom Lumens packages available, please consult factory. Values are within industry standard tolerances and not DLC listed.	**LL** - Less Lens[2] **FAC** - Flat Acrylic Clear General Distribution **FAD** - Flat Acrylic Diffuse General Distribution **FPC** - Flat Polycarbonate Clear General Distribution **FPD** - Flat Polycarbonate Diffuse General Distribution **NAC** - Narrow Acrylic Clear Distribution (Aisle) **CAC** - Curved Acrylic Clear General Distribution **CAD** - Curved Acrylic Diffuse General Distribution	**D** - 0 Degrees - Downlight (Standard) **3** - 30 Degrees - Wide[3] **6** - 60 Degrees - Extra Wide[3] **9** - 90 Degrees - Side Light[3] **I** - 180 Degrees - Indirect (100% Uplight)[2] *Each optic now is rotatable in 90 degree increments. **120, 150 degree presets available as special order. Contact factory for all other customized factory presets ***For 3, 6, or 9 presets, only the outside rows are factory set	**UNV** - 120V-277V **HV** - 347V- 480V	**840** - 80CRI/4000K **850** - 80CRI/5000K	**WHT** - White **BLK** - Black **MSV** - Metallic Silver

Controls	EM Options	Options
ALSC[1] - Airlink Synapse Control **ALSCSO2**[1] - Airlink Synapse Control System with 12-20' Motion Sensor **ALSCSO4**[1] - Airlink Synapse Control System with 20-40' Motion Sensor **ALBC** - AirLink Blue Controller **ALBCS2** - AirLink Blue Wireless Motion & Photo Sensor Controller (25-40' mounting height) **IMSBT2L** - Integral Bluetooth PIR Motion And Photo Sensor up to 40' **OMS2** - External PIR Sensor up to 40' **OMS2-480** - External PIR Sensor up to 40' for 480V applications *For low mounting height below 20ft, substitute "2" in nomenclature with "1" **For programming, please see "Sensors Sequence of Operations Charts"	**EM** - Externally Side-Mounted Battery Backup (Wattage Varies By Lumen Package)[4] **EMHV** - Externally Side-Mounted 30W HV Battery Backup for Maximum Egress Lumens *Refer EM lumen output table for light output **For cable mount applications with EM, it is recommended to use the TCH10 toggle kit to balance the fixture horizontally	**Standard** - 6" Dimming-Capable 5 Wire Leads **EXT** - Extended Leads (36" Dimming-Capable 5 Wire Leads) on END **CLDS** - Center Leads (36" Dimming-Capable 5 Wire Leads) on CENTER **E6CN** - 6ft Cord 16/3 Non-Dimming; No Plug Installed on TOP END **E6CD** - 6ft Cord 16/5 Dimming; No Plug Installed on TOP END **E10CN** - 10ft Cord 16/3 Non-Dimming; No Plug Installed on TOP END **E10CD** - 10ft Cord 16/5 Dimming; No Plug Installed on TOP END **C6CN** - 6ft Cord 16/3 Non-Dimming; No Plug Installed on TOP CENTER **C6CD** - 6ft Cord 16/5 Dimming; No Plug Installed on TOP CENTER **C10CN** - 10ft Cord 16/3 Non-Dimming; No Plug Installed on TOP CENTER **C10CD** - 10ft Cord 16/5 Dimming; No Plug Installed on TOP CENTER *See Cord Notes for specific applications **515P** - 5-15 Plug only (120V) **L515P** - 5-15P Twistlock Plug (125V) **L615P** - 6-15P Twistlock Plug (250V) **L715P** - 7-15P Twistlock Plug (277V) **L820P** - 8-20P Twistlock Plug (480V) **L2420P** - 24-20P Twistlock Plug only 347V) **SP1** - 10kV Surge Protection **JP** - Job Pack[6] (For Description See Product Dimensions Section)

Need more information?
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ACCESSORY ORDERING INFORMATION

Part Number	Order Number	Description
572488	SHC	Single Hub Chain Mount
783379R4	MCH10	10' Mini Hook Cable Hanger (2 per) - Up/Down adjustable and Left/Right adjustable
431373	TCH10	10' Toggle Hanger (2 Per) - Up/Down and Left/Right adjustable
767710	ACH	Single Hub/Aircraft Cable Mount
813349	PSHB PMK	Pendant Mount J-Box Kit
813352	PSHB SMK	Surface Mount Kit

Note: See product dimensions section for details.

LENS KIT ACCESSORY ORDERING INFORMATION

Part Number	Order Number	Description
816570	PSHB-LK-CAD-22	PSHB Lens Kit Curved Acrylic Diffuse 2ft-2 Row
816571	PSHB-LK-CAD-24	PSHB Lens Kit Curved Acrylic Diffuse 2ft-4 Row
816572	PSHB-LK-FPD-22	PSHB Lens Kit Flat Polycarbonate Diffuse 2ft-2 Row
816573	PSHB-LK-FPD-24	PSHB Lens Kit Flat Polycarbonate Diffuse 2ft-4 Row
816574	PSHB-LK-FAD-22	PSHB Lens Kit Flat Acrylic Diffuse 2ft-2 Row
816575	PSHB-LK-FAD-24	PSHB Lens Kit Flat Acrylic Diffuse 2ft-4 Row
816576	PSHB-LK-NAC-22	PSHB Lens Kit Narrow Acrylic Clear 2ft-2 Row
816577	PSHB-LK-NAC-24	PSHB Lens Kit Narrow Acrylic Clear 2ft-4 Row

1 Not available with 2ft fixture
2 LL not available with rotated optics greater than 90 degrees
3 For a 4 row configuration, only the 2 outside rows are rotated in the factory
4 For specific EM wattage requirements, please contact factory
5 EM/EMHV not available with ALSC controls options
6 EM units excluded from Job Packs and individually packaged

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Fig A4. Grow-light casing specifications sheet 2

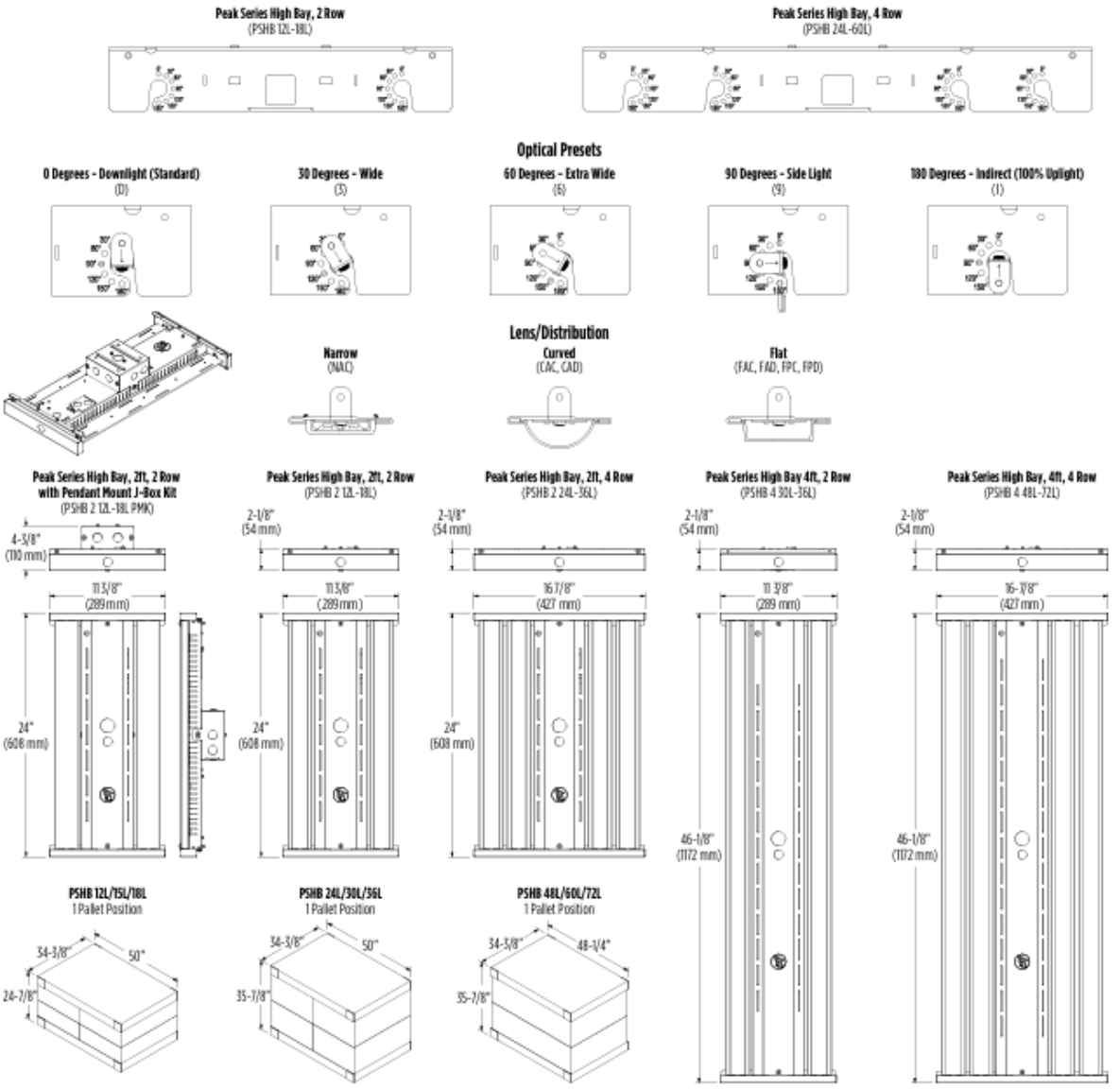
Peak Series High Bay (PSHB) Industrial

Type : _____

Have questions? Call us at (800) 436-7800

PRODUCT DIMENSIONS

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Shipping Weights - lbs (kg)	
PSHB2 12L-18L	9.6 (4.3)
PSHB2 24L-36L	14.4 (6.5)
PSHB4 48L-72L	27.8 (12.6)
EM	Add 4.2 (1.9)

Pallet Quantity				
Length (ft)	# of Rows	Lumen Package	Individual Box	Job Pack
2	2	12L/15L/18L	36	48
2	4	24L/30L/36L	36	48
4	2	48L/60L/72L	18	24

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Fig A5. Grow-light casing specifications sheet 3